

BAGGING, WITH DECISION TREES ONLY

Random Forest Classification and Regression

Many decision trees, each trained on a different part of the data. Their answers are combined into one, and the overfitting problem of a single tree goes away.

Random Forest = Bagging Where Every Model Is a Tree

BAGGING IN GENERAL

Base learners can be any algorithm. Decision tree, logistic regression, Naive Bayes, or a mix of all of them.

Trained in parallel, each on its own sample.
Answers combined at the end.

RANDOM FOREST

Same idea, one restriction: **every base learner is a decision tree**.

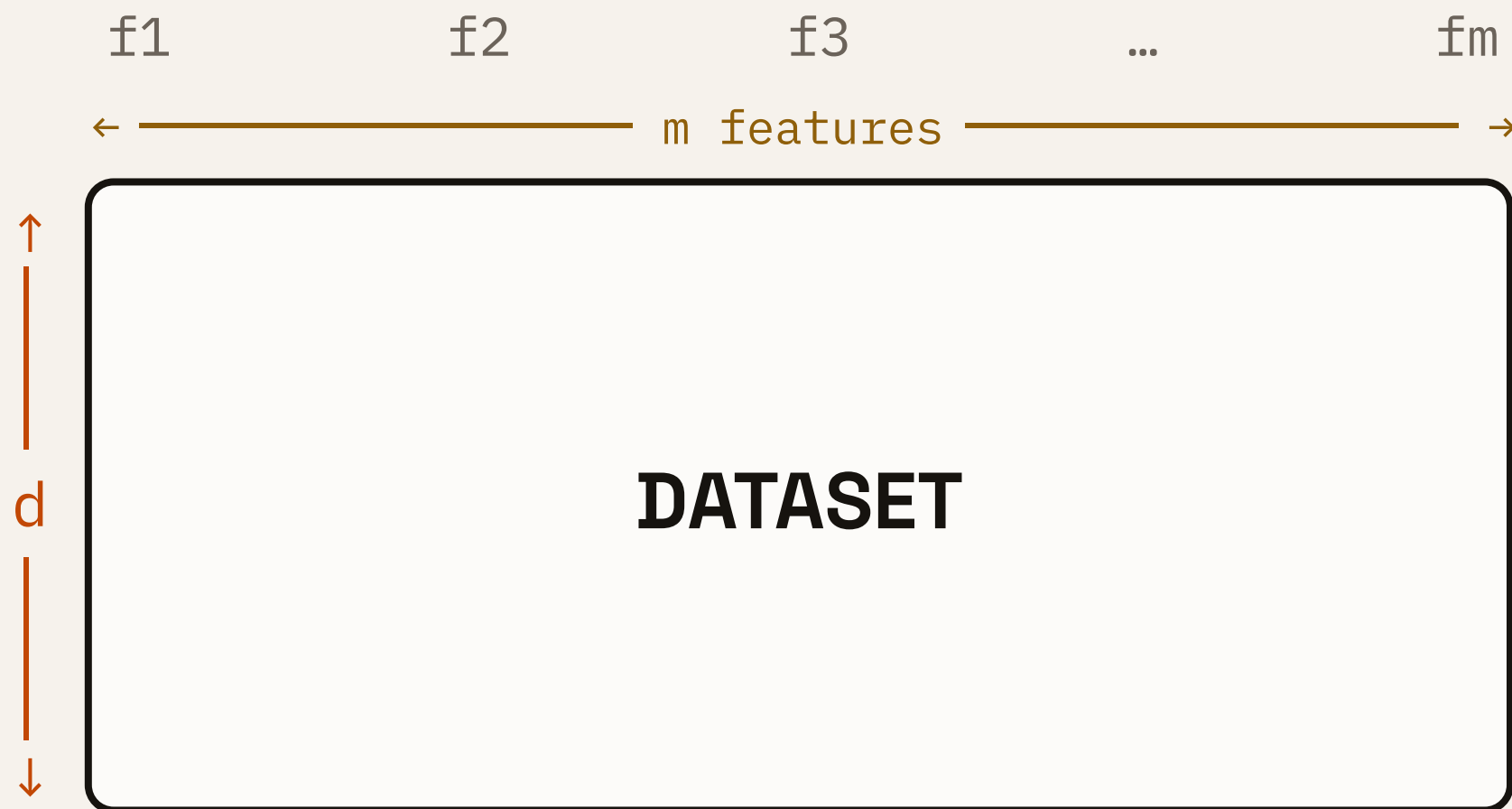
Many trees together, so we call it a forest.

There is no new tree maths here. Entropy and Gini for classification, variance for regression, exactly as before.

The only new thing to learn: how each tree gets its own version of the data.

NOTATION

d Rows, m Features



d how many rows the dataset has

m how many features it has, named f_1 to f_m

$d' < d$ Each tree is given d' rows, which is fewer than the full d . No tree ever sees the whole dataset.

Row Sampling and Feature Sampling

RS Row sampling

Pick some of the rows, not all of them.

1000 rows in the dataset → tree 1 gets
600 of them

+

FS Feature sampling

Pick some of the features, not all of them.

features $f_1 \dots f_m$ → tree 1 gets $\{f_1, f_2, f_4\}$

Tree 2 gets different rows and maybe $\{f_1, f_2, f_5, f_6\}$. Tree 3 gets something else again.

Different data means **genuinely different trees**.
Identical trees would all make the same mistake.

■ THE WORD TO REMEMBER

Sampling Happens With Replacement

WHAT IT MEANS

A row that has been picked is put back. It can be picked again for another tree, or even twice inside the same sample.

SO ROWS REPEAT

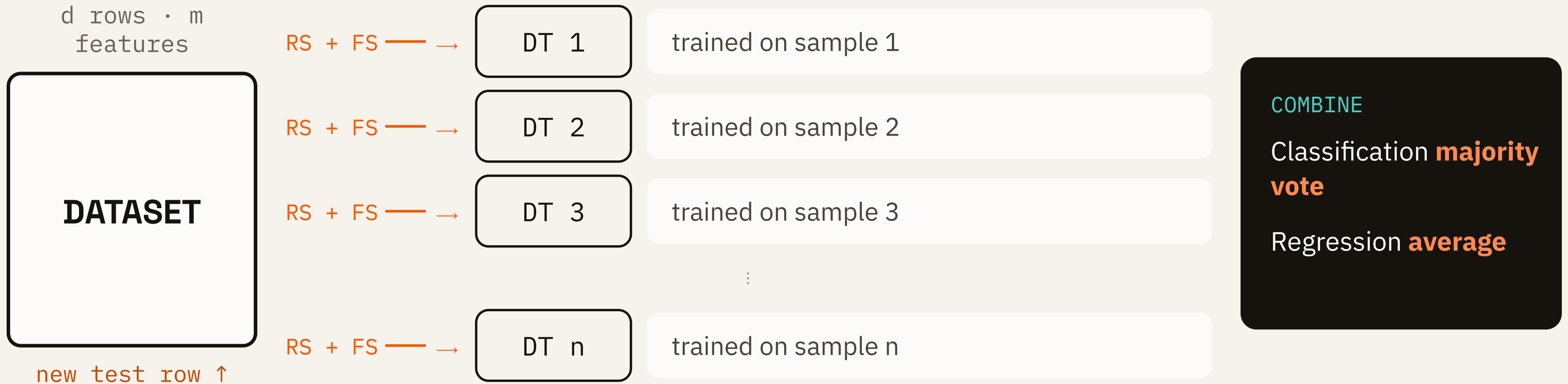
Some rows and some features do show up in more than one tree. That is fine and expected.

WHAT MUST NOT REPEAT

The **whole sample** . Most of the rows and features change from one tree to the next.

Without replacement we would run out of rows fast, and the trees would end up too small and too alike.

One Dataset, n Trees, One Answer



All trees train in parallel. Each tree is built with the rules we already know: entropy or Gini for classification, variance for regression.

Majority Voting Classifier

ONE NEW TEST ROW → EVERY TREE

DT 1 → 0

DT 2 → 0

DT 3 → 0

DT n → 1

Three votes for 0, one for 1 → final answer 0

Works for two classes and for many classes.
The class with the most votes wins.

Tree n was wrong here, and it did not matter.
It got outvoted by the trees that knew this
kind of row.

This is the single biggest practical benefit of a
forest: one bad tree cannot spoil the answer.

Average of the Tree Outputs

PREDICTING AN EXAM SCORE

DT 1 → 72

DT 2 → 69

DT 3 → 75

DT 4 → 76

$$(72 + 69 + 75 + 76) / 4 = 292 / 4$$

Predicted score 73

The outputs are numbers, so there is nothing to vote on. We add them and divide by the number of trees.

Each tree here is a **regression tree** : it splits using variance and each leaf holds the mean of its rows.

Averaging also smooths the answer. A single regression tree gives jumpy step predictions, a forest of them gives something much smoother.

COMMON INTERVIEW QUESTION

Why Use a Random Forest Instead of One Decision Tree?

A SINGLE DECISION TREE

With default settings it keeps splitting until the leaves are pure. It grows to its full depth.

Training accuracy is very high. Test accuracy drops. That is **overfitting**.

Unless we do pre-pruning or post-pruning, this is what happens by default.

SAID IN BIAS AND VARIANCE

Train accuracy high → **low bias**

Test accuracy low → **high variance**

A good, generalised model needs low bias *and* low variance. A single tree only gives us the first half.

THE ANSWER

Keep the Low Bias, Fix the High Variance

DECISION TREE

Low bias

High variance

→

RANDOM FOREST

Low bias

Low variance

The bias stays where it was. Only the variance comes down, and that is exactly the half we needed to fix.

Each tree was trained on **different rows and different features**, so each one overfits to a different corner of the data.

When we vote or average, those different mistakes cancel each other out instead of adding up.

One overfitted tree can ruin a prediction. A hundred overfitted trees, each overfitted differently, cannot.

ANOTHER WAY TO SEE IT

Add 200 New Rows. Does the Model Wobble?

ONE DECISION TREE

It can. A tree is built greedily from the top, so new data can change a split near the root.

Change the root and the whole shape below it changes too.

A FOREST OF TREES

The 200 new rows get spread across many samples, so each tree sees only a few of them.

No single tree changes much, and after voting or averaging the final prediction barely moves.

This steadiness is what people mean by calling Random Forest a robust model.

Better Accuracy, More Training Time

THE COST

Building 100 trees takes far longer than building 1. Training time and memory both go up.

Prediction also runs every tree before combining the answers.

Since the trees are independent, they can be trained in parallel across CPU cores.

SETTINGS YOU WILL ACTUALLY USE

<code>n_estimators</code>	how many trees. Default 100.
<code>max_features</code>	how many features a tree may look at per split.
<code>max_depth</code>	how deep each tree may grow.
<code>min_samples_split</code>	rows needed before a node may split.
<code>min_samples_leaf</code>	smallest number of rows allowed in a leaf.
<code>n_jobs = -1</code>	use all CPU cores to train in parallel.

The last four are the same pre-pruning settings from the previous session. In a forest they apply to every tree.

QUESTION

Does Every Tree Get the Same Number of Rows?

SHORT ANSWER

Yes, same size. What changes is *which* rows are inside.

By default each sample is as big as the original dataset. 1000 rows in, so every tree also gets 1000 rows, drawn with replacement.

```
max_samples = 0.6 → every tree gets 600  
rows instead
```

WHAT REPLACEMENT DOES INSIDE

Some rows get picked two or three times. Other rows never get picked at all.

```
sample for DT1 → 1000 rows long but only  
about 630 different rows about 370 rows left  
out
```

The left-out rows are called **out-of-bag** rows. That tree has never seen them, so they work as a free test set.

Setting `oob_score = True` tells scikit-learn to score the forest on those left-out rows. A free accuracy check, with no separate test split needed.

■ QUESTION

Are All 100 Trees Grown to Full Depth?

BY DEFAULT

Yes. Every tree splits until its leaves are pure, just like a single tree with default settings.

So every individual tree in the forest is overfitted.

AND THAT IS ON PURPOSE

Each tree is overfitted to a **different** sample, so they disagree about their mistakes.

Voting and averaging wipe out those disagreements. That is where the low variance comes from.

CAREFUL WITH HEAVY PRUNING

Prune each tree too hard and the trees turn shallow and start looking alike.

Similar trees make the same mistake, and then there is nothing left to cancel out.

For a single tree, pruning is what stops overfitting. For a forest, the voting does that job, so the trees are allowed to grow deep.

QUESTION

Can We Prune a Forest, and Are the Settings Shared?

PRE-PRUNING · USUALLY PREFERRED

Set the limits before training, so no tree ever grows past them.

`max_depth` · `min_samples_split` ·
`min_samples_leaf`

POST-PRUNING · POSSIBLE, RARELY USED

Grow the full tree, then cut branches back with `ccp_alpha`.

With 100 trees this is expensive: grow all of them fully, then prune all of them.

ARE THE SETTINGS PER TREE?

No. One setting, all trees.

You set the values once on the forest, and every tree inside is built with exactly those values.

You cannot give tree 1 a depth of 5 and tree 2 a depth of 12.

The trees differ because their **data** differs, never because their settings differ.

In practice most people leave the depth free and tune `n_estimators` and `max_features` first. Pre-pruning comes in only when training is too slow or the dataset is noisy.

■ RECAP

Five Things to Carry Forward

- 1 Random Forest is bagging where every base learner is a decision tree.
- 2 Each tree gets its own data: row sampling with replacement, plus feature sampling.
- 3 All the trees train in parallel, and no tree ever sees the full dataset.
- 4 Classification takes the majority vote. Regression takes the average.
- 5 A single tree gives low bias but high variance. A forest keeps the low bias and brings the variance down.

Next: the boosting side, starting with AdaBoost.